10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Review Exercise 6 — Vectors in the Plane

Unit 6 | Review + MCQs | New Book (PECTAA)

Review Exercise 6 (Solved)

Unit 6 is about vectors in the plane: coordinates of a point, xi + yj form, magnitude, unit vector, addition, subtraction and translation of figures.

1. Four possible answers are given for the following questions. Choose the correct answer.

(i) x-axis and y-axis divide a coordinate plane into _____ parts.

(a) one (b) two (c) three (d) four

Answer: (d) four — the two axes make four quadrants

(ii) P(4, −4) lies in _____ quadrant.

(a) first (b) second (c) third (d) fourth

Answer: (d) fourth — x is positive and y is negative, so the point is in the fourth quadrant

(iii) A vector having magnitude 1 is called:

(a) equal vector (b) parallel vector (c) unit vector (d) zero vector

Answer: (c) unit vector — a unit vector has length exactly 1

(iv) What is the value of |3i + 4j|?

(a) 3 (b) 4 (c) 5 (d) 7

Answer: (c) 5 — $\sqrt{(9 + 16)} = \sqrt{25} = 5$

(v) If a = λb, then a and b are:

(a) equal (b) parallel (c) perpendicular (d) non-parallel

Answer: (b) parallel — one vector is a scalar multiple of the other

(vi) If OA = a and OB = b, then AB is:

(a) b − a (b) a − b (c) a + b (d) b + a

Answer: (a) b − a — AB = OB − OA = b − a

(vii) Translation vector shows:

(a) deformation (b) rotation (c) movement (d) enlargement

Answer: (c) movement — every point slides by the same amount, without turning

(viii) Sum of two vectors is:

(a) a triangle (b) a vector (c) a scalar (d) a length

Answer: (b) a vector — the sum of two vectors is again a vector (the resultant)

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(ix) The position vector of point P(3, −2) with respect to O is:

(a) 3i + 2j (b) 3i − 2j (c) −3i + 2j (d) 2i − 3j

Answer: (b) 3i − 2j — OP = x i + y j = 3i − 2j

(x) Vector from point P(3, 4) to origin is:

(a) 3i + 4j (b) −3i + 4j (c) −3i − 4j (d) 3i − 4j

Answer: (c) −3i − 4j — PO = O − P = −3i − 4j

2. Find magnitude of the AB.

$\overline{AB} = (x_2 - x_1)\bar{i} + (y_2 - y_1)\bar{j}$

$|\overline{AB}| = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

(i) A(7, 7), B(−12, 0)

$\overline{AB} = (-12 - 7)\bar{i} + (0 - 7)\bar{j} = -19\bar{i} - 7\bar{j}$

$|\overline{AB}| = \sqrt{((-19)^2 + (-7)^2)} = \sqrt{(361 + 49)}$

Answer: |AB| = √410

(ii) A(9, 3), B(2, 11)

$\overline{AB} = (2 - 9)\bar{i} + (11 - 3)\bar{j} = -7\bar{i} + 8\bar{j}$

$|\overline{AB}| = \sqrt{((-7)^2 + (8)^2)} = \sqrt{(49 + 64)}$

Answer: |AB| = √113

3. Find a unit vector in the direction of a = (5/3)i + (1/3)j.

$|a| = \sqrt{((\frac{5}{3})^2 + (\frac{1}{3})^2)}$

$= \sqrt{(\frac{25}{9} + \frac{1}{9})} = \sqrt{(\frac{26}{9})} = \frac{\sqrt{26}}{3}$

$\hat{a} = (\frac{5}{3} \bar{i} + \frac{1}{3} \bar{j}) \div \frac{\sqrt{26}}{3}$

$= \frac{5}{3} \times \frac{3}{\sqrt{26}} \bar{i} + \frac{1}{3} \times \frac{3}{\sqrt{26}} \bar{j}$

$= \frac{5}{\sqrt{26}} \bar{i} + \frac{1}{\sqrt{26}} \bar{j}$

Answer: â = (5/√26)i + (1/√26)j

4. If a = 2i − j, b = 3i + j and c = 4i + j, then find the following vectors.

(i) 5b − a + c (ii) 8a + b + 5c (iii) c + b − 4a

(i) 5b − a + c

$= 5(3\bar{i} + \bar{j}) - (2\bar{i} - \bar{j}) + (4\bar{i} + \bar{j})$

$= (15 - 2 + 4)\bar{i} + (5 + 1 + 1)\bar{j}$

Answer: $5b - a + c = 17\bar{i} + 7\bar{j}$

(ii) 8a + b + 5c

$= 8(2\bar{i} - \bar{j}) + (3\bar{i} + \bar{j}) + 5(4\bar{i} + \bar{j})$

$= (16 + 3 + 20)\bar{i} + (-8 + 1 + 5)\bar{j}$

Answer: $8a + b + 5c = 39\bar{i} - 2\bar{j}$

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(iii) c + b − 4a

$= (4\bar{i} + \bar{j}) + (3\bar{i} + \bar{j}) - 4(2\bar{i} - \bar{j})$

$= (4 + 3 - 8)\bar{i} + (1 + 1 + 4)\bar{j}$

Answer: $\mathbf{c + b - 4a = -\bar{i} + 6\bar{j}}$

5. Find the values of x and y in the following equation.

$(2x\bar{i} + y\bar{j}) + (-\bar{i} + 5\bar{j}) = \frac{1}{4}\bar{i} - 8\bar{j}$

$(2x - 1)\bar{i} + (y + 5)\bar{j} = \frac{1}{4}\bar{i} - 8\bar{j}$

Comparing the i parts: $2x - 1 = 1/4$

$2x = 1/4 + 1$

$2x = 5/4$

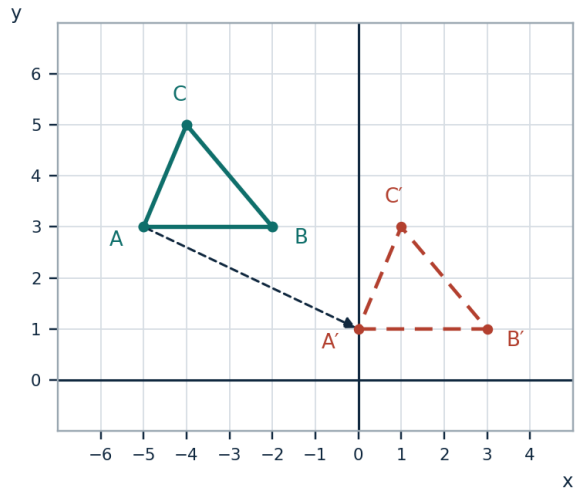
$x = 5/8$

Comparing the j parts: $y + 5 = -8$

$y = -13$

Answer: x = 5/8, y = −13

6. Plot A(−5, 3), B(−2, 3) and C(−4, 5) to form a triangle ABC. Also translate ABC to A′B′C′ by the translation vector 5i − 2j.



ABC → A′B′C′ (5i − 2j)

Add 5 to x, −2 to y

A′ = (−5 + 5)i + (3 − 2)j

= 0i + j = (0, 1)

B′ = (−2 + 5)i + (3 − 2)j = (3, 1)

C′ = (−4 + 5)i + (5 − 2)j = (1, 3)

Answer: A′(0, 1), B′(3, 1), C′(1, 3)

7. Use vectors to show that ABCD is a parallelogram, where the points are A(2, 3), B(6, 3), C(7, 6) and D(3, 6).

$\overline{AB} = (6 - 2)\bar{i} + (3 - 3)\bar{j} = 4\bar{i}$

$\overline{DC} = (7 - 3)\bar{i} + (6 - 6)\bar{j} = 4\bar{i}$

$\overline{BC} = (7 - 6)\bar{i} + (6 - 3)\bar{j} = \bar{i} + 3\bar{j}$

$\overline{AD} = (3 - 2)\bar{i} + (6 - 3)\bar{j} = \bar{i} + 3\bar{j}$

AB = DC and BC = AD, so both pairs of opposite sides are equal and parallel.

Answer: ABCD is a parallelogram

8. Use vectors to show that triangle ABC is an isosceles triangle, where the points A, B and C have coordinates (1, 2), (4, 6) and (7, 2) respectively.

$\overline{AB} = (4 - 1)\bar{i} + (6 - 2)\bar{j} = 3\bar{i} + 4\bar{j}$

$\overline{BC} = (7 - 4)\bar{i} + (2 - 6)\bar{j} = 3\bar{i} - 4\bar{j}$

$\overline{AC} = (7 - 1)\bar{i} + (2 - 2)\bar{j} = 6\bar{i}$

$|\overline{AB}| = \sqrt{(9 + 16)} = 5, \quad |\overline{BC}| = \sqrt{(9 + 16)} = 5, \quad |\overline{AC}| = 6$

AB and BC are equal in length, so the triangle is isosceles.

Answer: |AB| = |BC| = 5, so ABC is isosceles

9. What is the magnitude of velocity v = 6i + 8j?

$|v| = \sqrt{((6)^2 + (8)^2)} = \sqrt{(36 + 64)} = \sqrt{100}$

Answer: |v| = 10

10. An aircraft is flying due east with an airspeed of 200 km/h. There is a wind blowing due north at 60 km/h. Find the resultant velocity and its magnitude.

Velocity of aircraft $= 200\bar{i}$, wind $= 60\bar{j}$

Resultant $= 200\bar{i} + 60\bar{j}$

$|\text{Resultant}| = \sqrt{((200)^2 + (60)^2)} = \sqrt{(40000 + 3600)}$

$= \sqrt{43600} = 20\sqrt{109}$

$= 208.8$ km/h (about 209 km/h)

Answer: Resultant velocity = 200i + 60j, magnitude ≈ 208.8 km/h

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